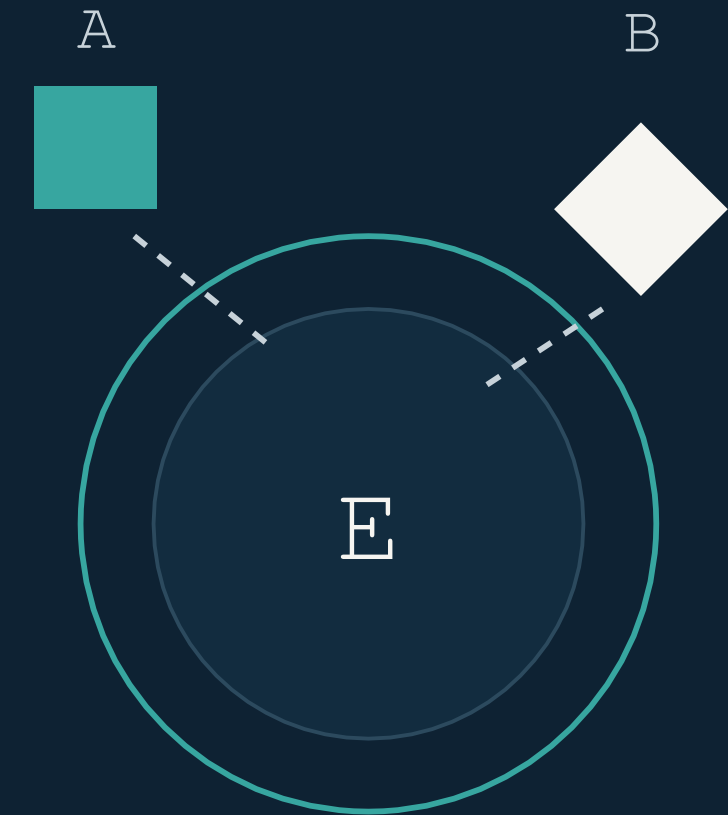


ENZYME MECHANISMS

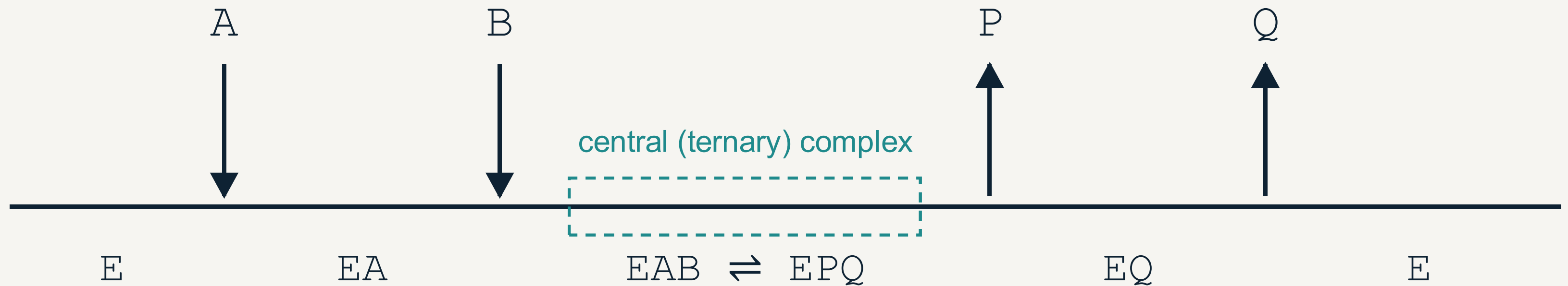
Bi Bi Enzyme Kinetics

Sequential vs. ping pong mechanisms
in two-substrate, two-product
reactions



SEQUENTIAL · CLELAND DIAGRAM

Ordered Bi Bi Mechanism



Substrates bind in a fixed order; both must be bound before catalysis or any product release occurs.

SEQUENTIAL · STEADY STATE

Ordered Bi Bi: Rate Equation Derivation



$$v = \frac{V_{\max} [A][B]}{K_{ia}K_b + K_a[B] + K_b[A] + [A][B]}$$

K_a — Michaelis constant for A

K_b — Michaelis constant for B

K_{ia} — dissociation constant of the EA complex

At saturating [B] the expression collapses to standard Michaelis-Menten kinetics in [A].

Worked Derivation: Ordered Bi Bi Rate Equation

- 1 Write the elementary steps



- 2 Apply steady state to each complex

$$d[EA]/dt = d[EAB]/dt = d[EPQ]/dt = d[EQ]/dt = 0$$

- 3 Conserve total enzyme

$$[E]_t = [E] + [EA] + [EAB] + [EPQ] + [EQ]$$

- 4 Solve for each enzyme fraction

weight every term with its King-Altman pattern

- 5 Substitute into the catalytic rate

$$v = k_3[EAB], \text{ then collect terms in } [A], [B] \text{ and } [A][B]$$

- 6 Final form

$$v = \frac{V_{\max} [A][B]}{K_{ia}K_b + K_a[B] + K_b[A] + [A][B]}$$

The extra $K_{ia}K_b$ term, absent in ping pong, reflects A binding before B —and it is what produces intersecting Lineweaver-Burk lines.

Real-World Example: Lactate Dehydrogenase

$E \rightarrow EA$

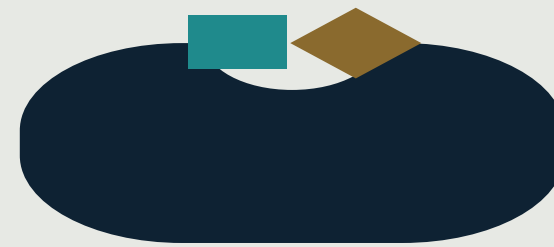
NADH binds first



The coenzyme is the obligatory leading substrate (A).

$EA \rightarrow EAB$

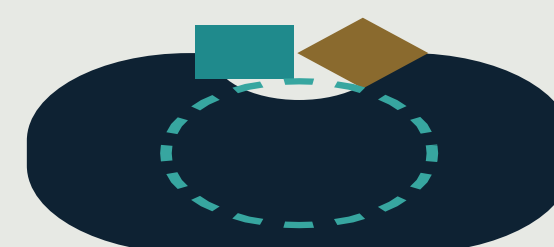
Pyruvate joins



Ternary complex
 $E \cdot \text{NADH} \cdot \text{pyruvate}$ forms.

$EAB \rightleftharpoons EPQ$

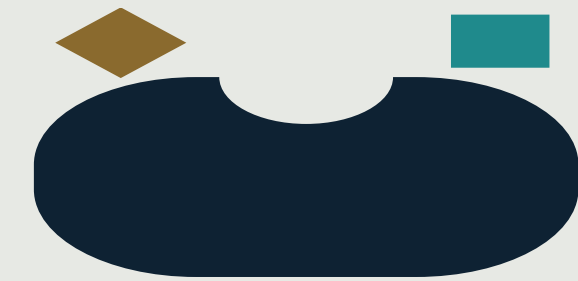
Hydride transfer



Chemistry happens only inside
the ternary complex.

$EPQ \rightarrow EQ \rightarrow E$

Ordered release



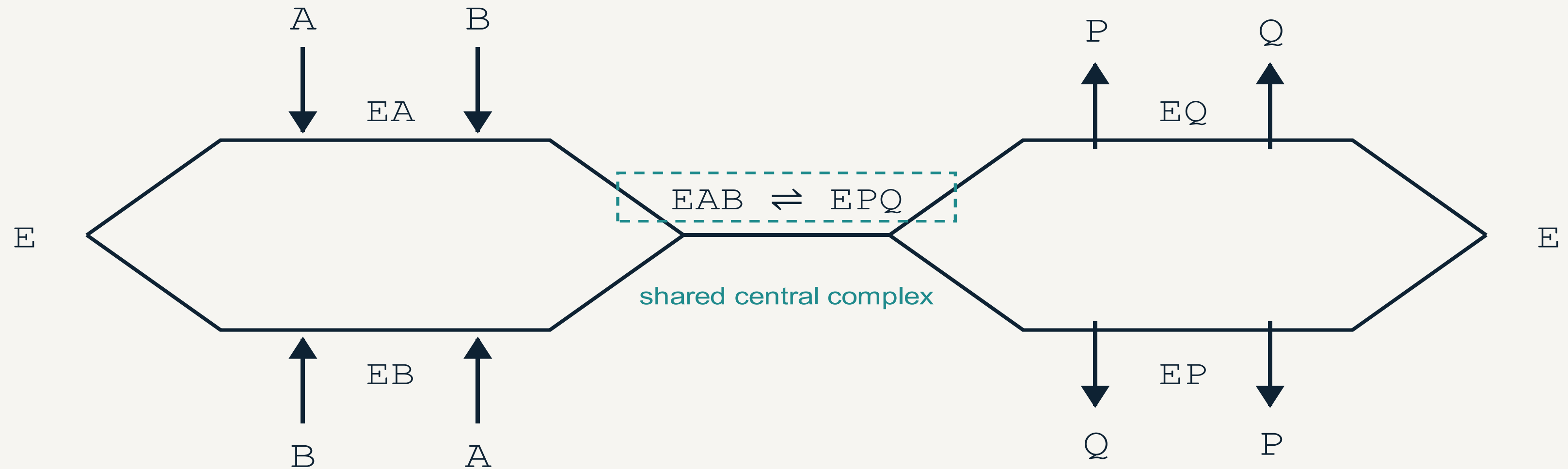
Lactate (P) leaves, then NAD^+ (Q) last.

$E \rightarrow EA \rightarrow EAB \rightleftharpoons EPQ \rightarrow EQ \rightarrow E$

NADH always binds before pyruvate, and release is ordered too —lactate before NAD^+ . Product inhibition studies confirm the sequence.

SEQUENTIAL, UNORDERED · CLELAND DIAGRAM

Random Bi Bi Mechanism



Either substrate can bind first — a ternary complex still forms before any product is released.

SEQUENTIAL · RAPID EQUILIBRIUM

Random Bi Bi: Rapid Equilibrium Derivation



Assumes A and B binding equilibrate much faster than the catalytic step.

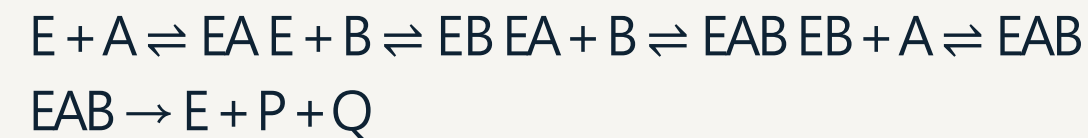
$$v = \frac{V_{\max} [A][B]}{K_{ia}K_b + K_b[A] + K_a[B] + [A][B]}$$

Algebraically this matches ordered Bi Bi, so random and ordered are separated by product inhibition patterns or isotope exchange — not by the shape of the rate equation.

RAPID EQUILIBRIUM

Worked Derivation: Random Bi Bi Rate Equation

- 1 Write the parallel equilibria plus catalysis



- 2 State the assumption

All binding steps equilibrate much faster than catalysis, so true dissociation constants (K_a , K_b , K_{ia} , K_{ib}) apply.

- 3 Write the dissociation constants

$$K_{ia} = [E][A]/[EA] \quad K_{ib} = [E][B]/[EB] \quad \text{thermodynamic box: } K_{ia} \cdot K_b = K_{ib} \cdot K_a$$

- 4 Conserve total enzyme

$$[E]_t = [E] + [EA] + [EB] + [EAB]$$

- 5 Solve for $[EAB]$ and substitute

$$v = k_{cat}[EAB]$$

- 6 Final form

$$v = \frac{V_{max} [A][B]}{K_{ia}K_b + K_b[A] + K_a[B] + [A][B]}$$

Identical in form to ordered Bi Bi: steady-state kinetics alone cannot separate the two, so product inhibition or isotope exchange is required.

Real-World Example: Creatine Kinase

$E \rightarrow E \cdot \text{ATP}$

ATP binds first

Substrate A takes the open site.



$E \cdot \text{ATP} \cdot \text{creatine}$

One shared ternary complex



Either binding order arrives here; phosphoryl transfer then follows.



P releases first

Phosphocreatine leaves

ADP follows, regenerating E .

$E \rightarrow E \cdot \text{creatine}$

Creatine binds first

Substrate B is equally free to lead.



Q releases first

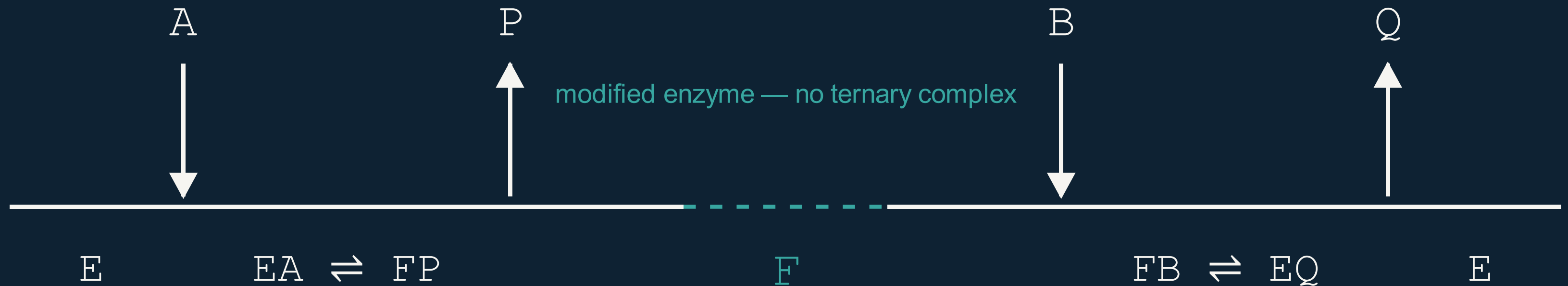
ADP leaves

Phosphocreatine follows.

A rapid-equilibrium random Bi Bi enzyme, and the engine of the phosphocreatine shuttle that buffers ATP in muscle and brain.

DOUBLE DISPLACEMENT · CLELAND DIAGRAM

Ping Pong Bi Bi Mechanism



The enzyme is chemically modified between binding events: P leaves before B arrives. Seen in transaminases and some kinases.

DOUBLE DISPLACEMENT · STEADY STATE

Ping Pong Bi Bi: Rate Equation Derivation



$$v = \frac{V_{\max} [A][B]}{K_a[B] + K_b[A] + [A][B]}$$

No $K_{ia}K_b$ term: with no ternary complex there is no simultaneous-binding equilibrium to describe.

That missing constant is why the double-reciprocal lines come out parallel.

Slope in $1/[A]$ is independent of $[B]$; only the intercept moves.

STEADY STATE

Worked Derivation: Ping Pong Bi Bi Rate Equation

- 1 Write the elementary steps



- 2 Apply the steady-state assumption

$$d[EA]/dt = 0 \quad d[FB]/dt = 0$$

- 3 Conserve total enzyme

$$[E]_t = [E] + [EA] + [F] + [FB]$$

- 4 Solve for the bound forms

$[EA]$ and $[FB]$ in terms of $[E]$, $[F]$, $[A]$, $[B]$

- 5 Equate the two half-reaction rates

$v = k_2[EA] = k_4[FB]$, eliminate $[E]$ and $[F]$

- 6 Final form

$$v = \frac{V_{\max} [A][B]}{K_a[B] + K_b[A] + [A][B]}$$

No constant term survives: the two half-reactions are independent, so nothing describes simultaneous binding.

Real-World Example: Serine Proteases and Esterases



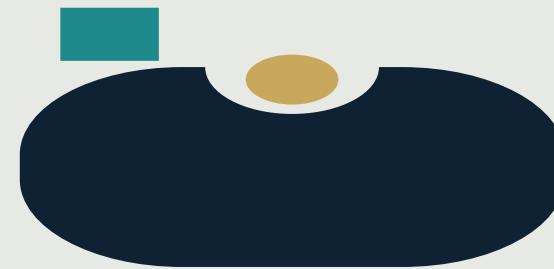
Substrate binds Ser



R-CO-X, where X = -OR' for an ester or -NH-R' for a peptide.



First product leaves



Alcohol or amine departs; the acyl-enzyme F remains.



Water attacks



H₂O is substrate B, entering the vacated site.



Enzyme regenerated



R-COOH leaves and free enzyme returns.

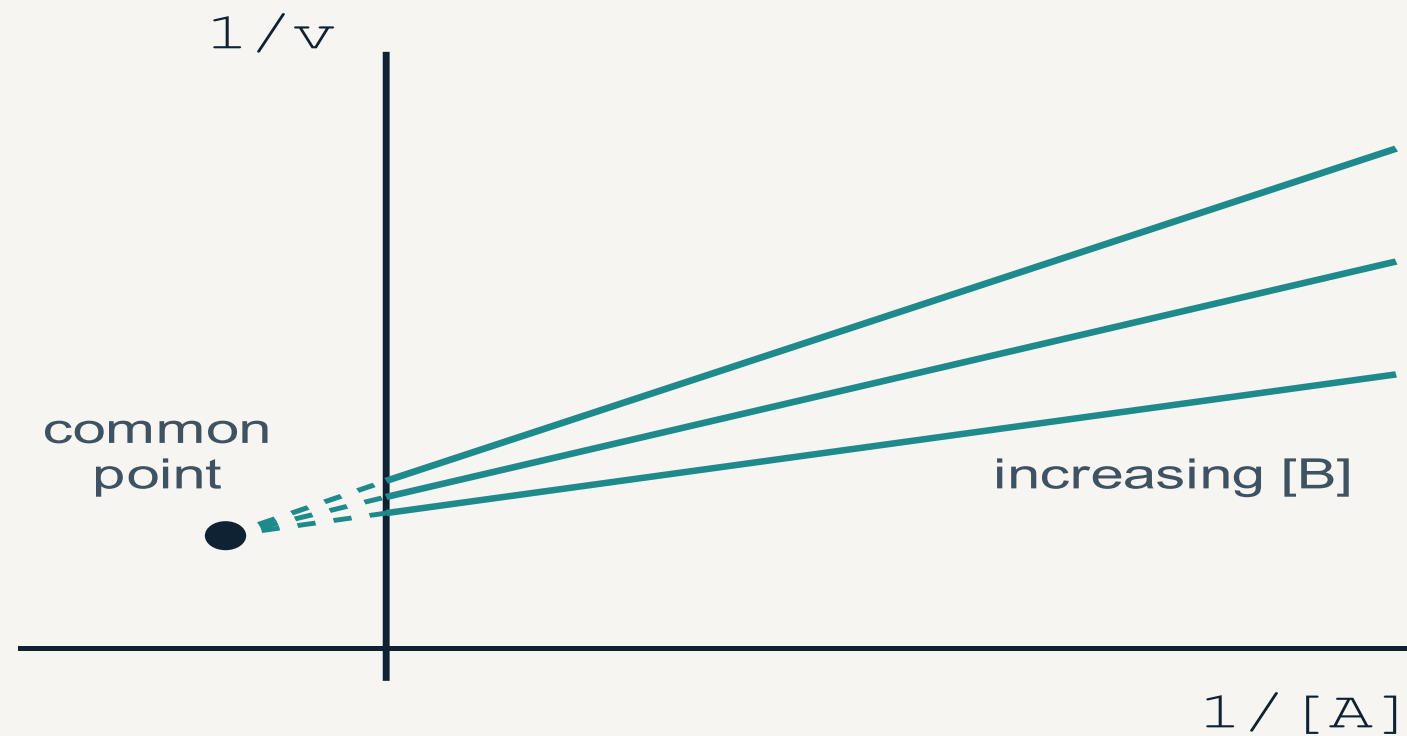


Acetylcholinesterase hydrolysing acetylcholine; chymotrypsin and trypsin cleaving peptide bonds—all run through a covalent acyl-enzyme.

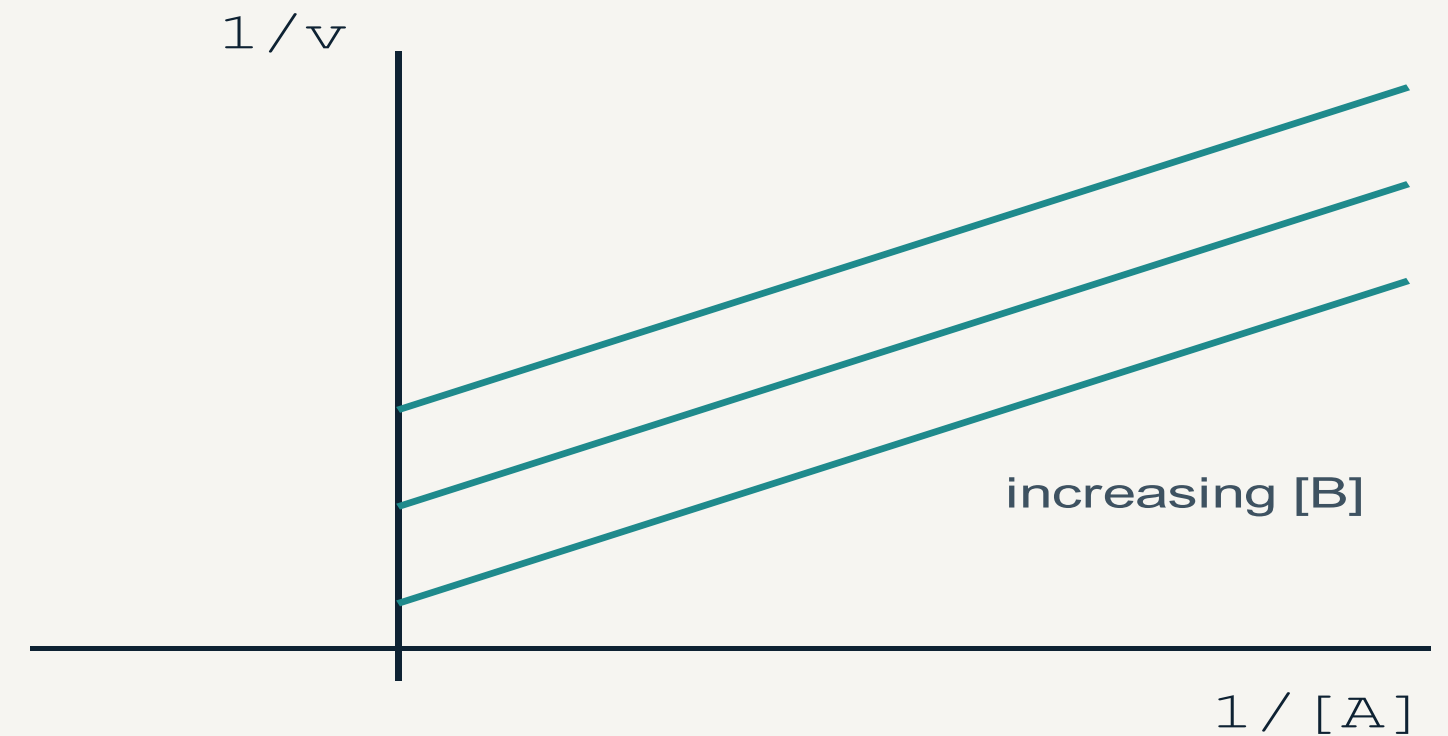
DOUBLE-RECIPROCAL PLOTS

Distinguishing the Mechanisms Kinetically

Sequential (ordered or random)



Ping pong



Intersecting lines indicate a ternary complex; parallel lines indicate a substituted-enzyme (ping pong) mechanism.

SUMMARY

Rate Equation Comparison

	Ordered Bi Bi	Random Bi Bi	Ping Pong Bi Bi
Denominator of v	$K_i a K_b + K_a [B] + K_b [A] + [A][B]$	$K_i a K_b + K_a [B] + K_b [A] + [A][B]$	$K_a [B] + K_b [A] + [A][B]$
Ternary complex	✓ yes	✓ yes	✗ no
Lineweaver-Burk	Intersecting	Intersecting	Parallel
Constant term	$K_i a K_b$	$K_i a K_b$	none

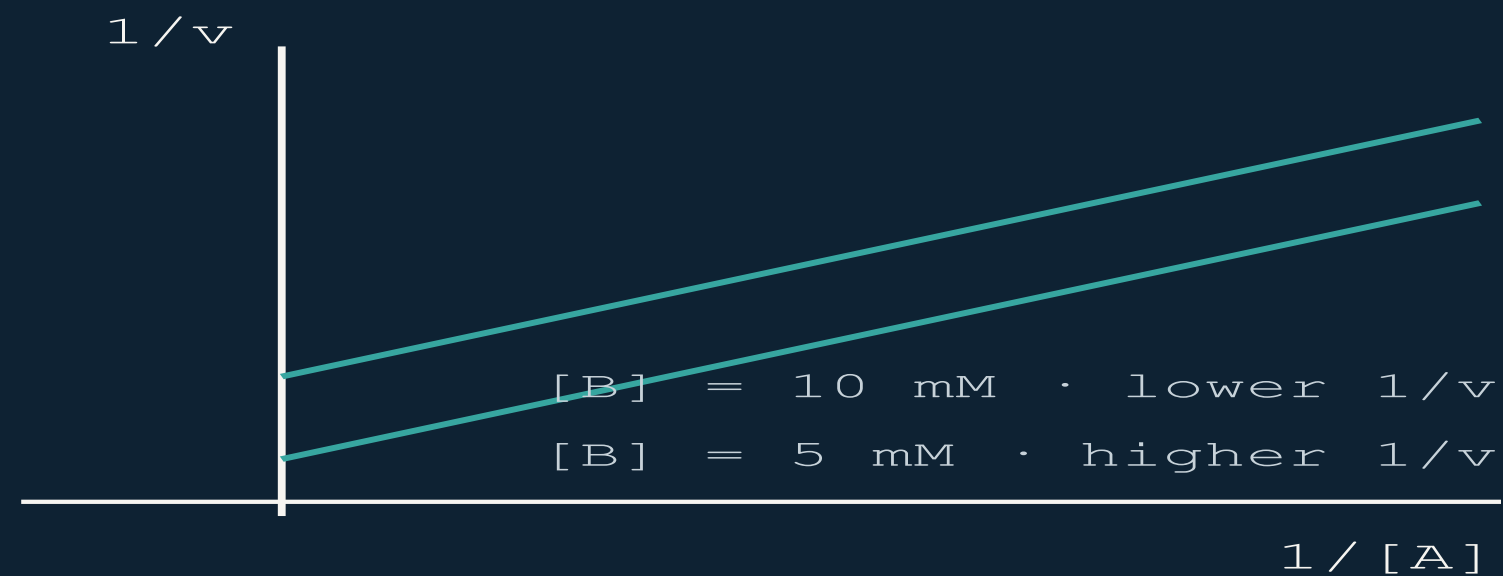
Problem 1: Identify the Mechanism



[A] (mM)	v at [B] = 5 mM	v at [B] = 10 mM
1	25.00	28.57
2	33.33	40.00
4	40.00	50.00
10	45.45	58.82

- 1. Calculate $1/v$ and $1/[A]$ for every point.
- 2. Sketch the double-reciprocal plot at each [B].
- 3. Which mechanism does the pattern indicate?
- 4. Determine K_a , K_b and V_{max} .

Solution 1: Ping Pong Bi Bi



- Slope = $K_a/V_{\max} = 0.02$ at both $[B]$
- Intercepts: 0.020 at $[B]=5$, 0.015 at $[B]=10$
- Secondary plot slope = $K_b/V_{\max} = 0.05$
- Secondary intercept = $1/V_{\max} = 0.010$

Parallel lines → ping pong Bi Bi: no ternary complex.

$$V_{\max} = 100 \mu\text{mol}/\text{min}/\text{mg} \quad K_a = 2 \text{ mM} \quad K_b = 5 \text{ mM}$$

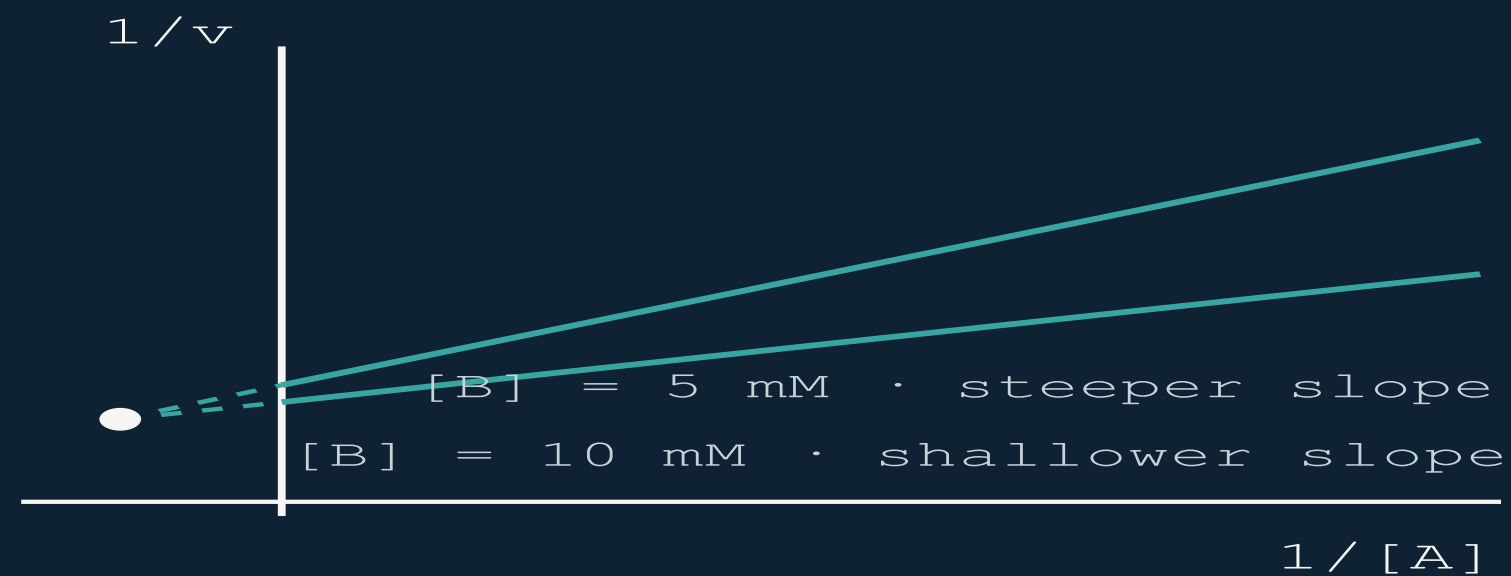
Problem 2: Identify the Mechanism



[A] (mM)	v at [B] = 5 mM	v at [B] = 10 mM
1	12.50	18.18
2	20.00	28.57
4	28.57	40.00
10	38.46	52.63

- 1. Calculate $1/v$ and $1/[A]$.
- 2. What pattern do the lines show?
- 3. What does that say about the mechanism?
- 4. Calculate V_{max} , K_a and K_b .

Solution 2: Sequential (Ordered) Bi Bi



Intersecting lines $\rightarrow$ ternary complex, sequential Bi Bi.

- $\rightarrow$ Slopes: 0.060 at $[B]=5$, 0.040 at $[B]=10$
- $\rightarrow$ Intercepts: 0.020 and 0.015
- $\rightarrow$ Secondary: $K_b/V_{\max} = 0.05$, $1/V_{\max} = 0.010$
- $\rightarrow$ Slope-of-slopes = $K_{ia} \cdot K_b / V_{\max} = 0.20 \rightarrow K_{ia} \cdot K_b = 20$
- $\rightarrow$ Intercept gives $K_a / V_{\max} = 0.020 \rightarrow K_a = 2 \text{ mM}$

$$V_{\max} = 100 \mu\text{mol}/\text{min}/\text{mg} \quad K_a = 2 \text{ mM} \quad K_b = 5 \text{ mM} \quad K_{ia} = 4 \text{ mM}$$

Problem 3: Direct Calculation

An enzyme follows ordered Bi Bi kinetics with $V_{\max} = 80$ $\mu\text{mol}/\text{min}/\text{mg}$, $K_a = 3$ mM, $K_b = 6$ mM and $K_{ia} = 2$ mM. Calculate v at $[A] = 6$ mM and $[B] = 12$ mM.

$$v = V_{\max}[A][B] / (K_{ia} \cdot K_b + K_a[B] + K_b[A] + [A][B])$$

Solution 3: Direct Calculation

$$\rightarrow K_{ia} \cdot K_b = 2 \times 6 = 12$$

$$\rightarrow K_a[B] = 3 \times 12 = 36$$

$$\rightarrow K_b[A] = 6 \times 6 = 36$$

$$\rightarrow [A][B] = 6 \times 12 = 72$$

$$\rightarrow \text{Denominator} = 12 + 36 + 36 + 72 = 156$$

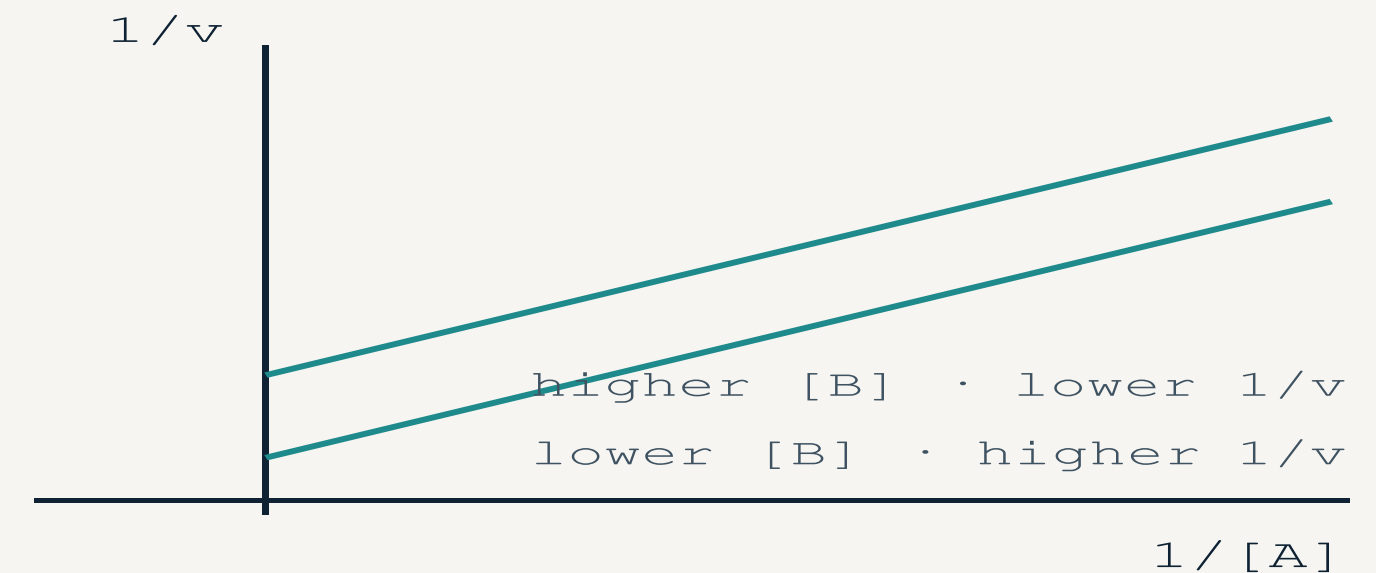
$$\rightarrow \text{Numerator} = 80 \times 6 \times 12 = 5760$$

$$v = 5760 / 156 = 36.9 \text{ } \mu\text{mol/min/mg}$$

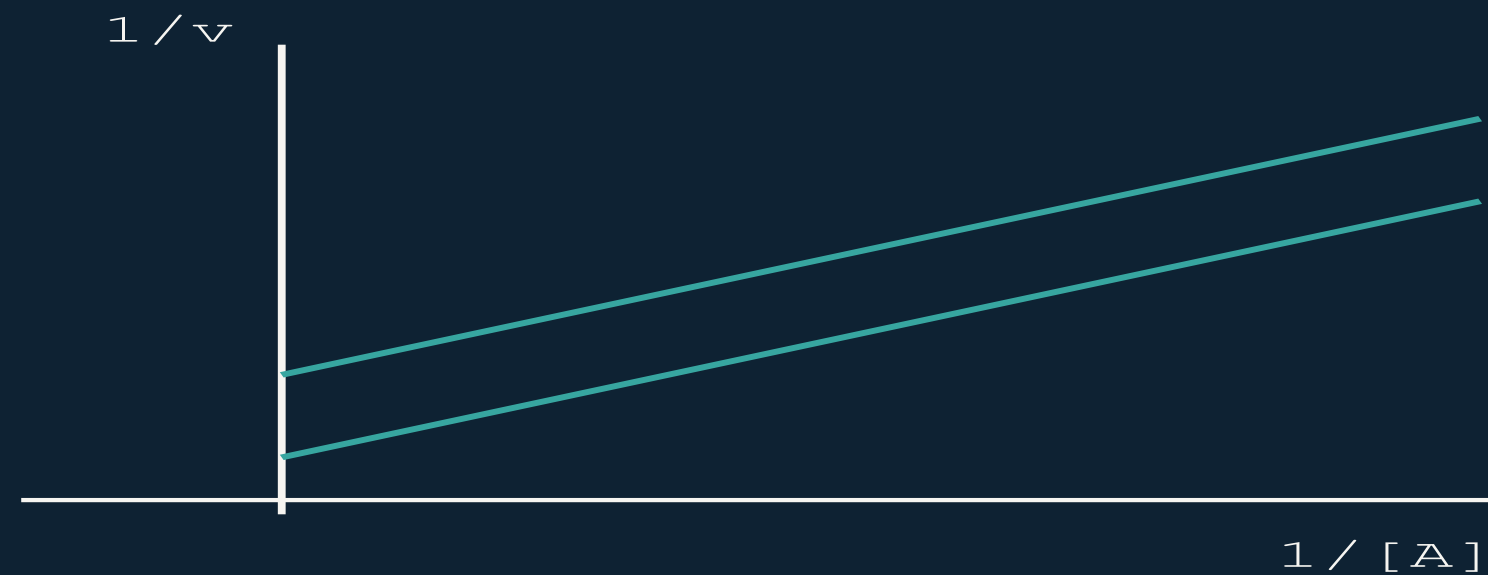
Problem 4: Conceptual Question

A student runs $1/v$ against $1/[A]$ at several fixed $[B]$ for an unknown Bi Bi enzyme. The lines come out parallel.

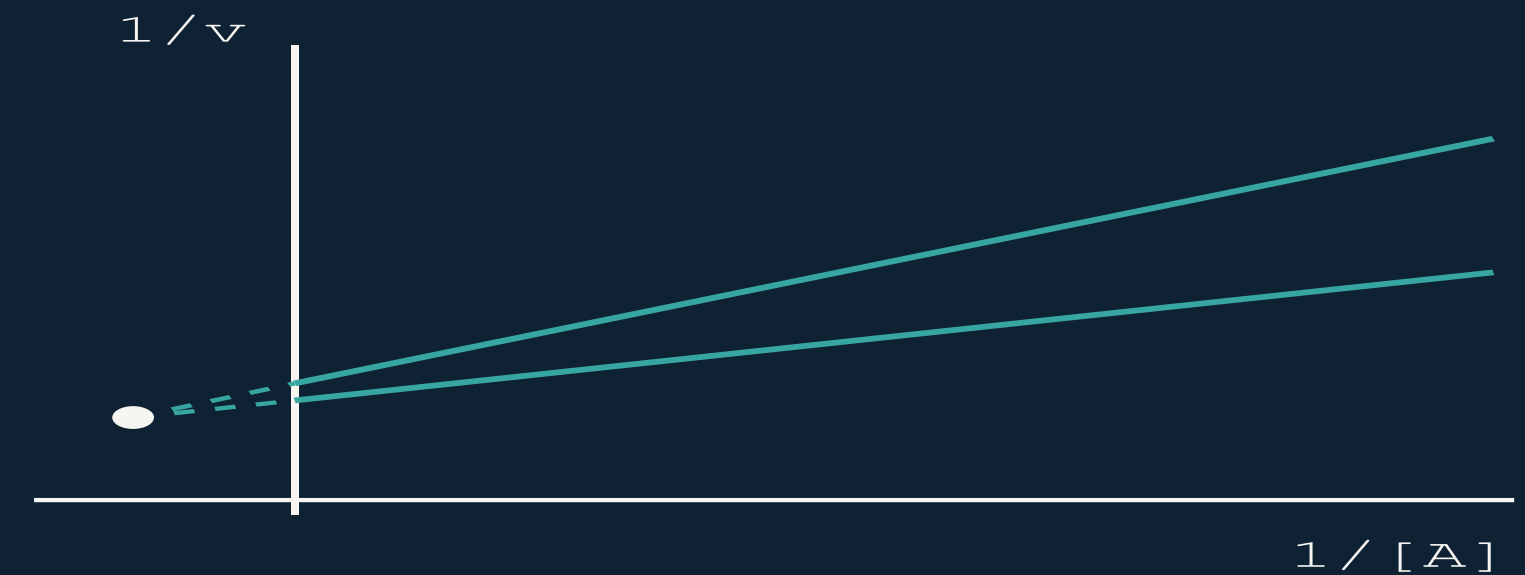
- a) Which mechanism is indicated?
- b) What would the plot look like for ordered Bi Bi instead?
- c) Name one real enzyme for each mechanism.



Solution 4: Conceptual Answer



Ping pong — parallel



Ordered — intersecting left of the axis

- Ping pong Bi Bi: no ternary complex, and the modified enzyme F carries the group between half-reactions.
- Ordered Bi Bi lines meet at a common point, because the $K_{ia}K_b$ term couples the two substrates.
- Ping pong: chymotrypsin or acetylcholinesterase. Ordered: lactate dehydrogenase.